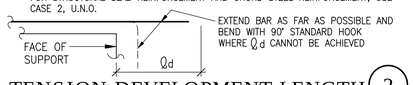
NOTES:
1. TABLE FOR USE WITH NORMAL WEIGHT HARDROCK CONCRETE AND GRADE 60 UNCOATED REINFORCING BARS. FOR LIGHTWEIGHT AGGREGATE USE 1.3ℓ
2. CLASS A – HALF OR LESS OF THE BARS ARE SPLICED WITHIN A REQUIRED LAP LENGTH. CLASS B – MORE THAN HALF OF THE BARS ARE SPLICED WITHIN A REQUIRED LAP LENGTH.
3. TOP BARS ARE HORIZONTAL BARS WITH 12" OR MORE OF CONCRETE CAST IN THE MEMBER BELOW THE BAR.
4. FOR BARS ENCLOSED IN STANDARD COLUMN SPIRALS, USE 0.75ℓ OR 12" MIN.
5. LAP SPLICES OF INDIVIDUAL BARS WITH A BUNDLE SHALL BE 1.2ℓ FOR THAT BAR IN A 3–BAR BUNDLE AND 1.33ℓ FOR A 4–BAR BUNDLE. ENTIRE BUNDLES SHALL NOT BE LAP SPLICED AT THE SAME LOCATION. SPLICES FOR INDIVIDUAL BARS WITHIN A BUNDLE SHALL BE STAGGERED SUCH THAT THEY DO NOT OVERLAP.
6. ℓ – BASIC LAP LENGTH, SHOWN AT LEFT.
7. CASE SELECTION
   –FOR FOUNDATION, SLAB ON GRADE, COLUMN REINFORCEMENT AND DOWELS, BEAM REINFORCEMENT, WALL REINFORCEMENT AND DOWELS, AND WALLS WITH A SINGLE MAT OF STEEL CENTERED IN THE WALL AND DOWELS USE CASE 1 U.N.O.
   –FOR STRUCTURAL SLAB REINFORCEMENT AND CHORD STEEL REINFORCEMENT, USE CASE 2, U.N.O.

BAR SIZE	LAP CLASS	f'c = 3000 psi			
		TOP BARS		OTHER BARS	
		CASE 1	CASE 2	CASE 1	CASE 2
#3	A	19	28	15	22
	B	24	36	19	28
#4	A	25	37	19	29
	B	32	48	25	37
#5	A	31	47	24	36
	B	40	60	31	47
#6	A	37	56	29	43
	B	48	72	37	56
#7	A	54	81	42	63
	B	70	106	54	81

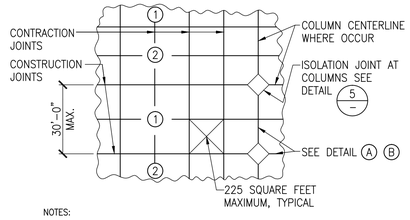
TENSION LAP SLICE (CONCRETE ONLY) 1
NTS

NOTES:
1. TABLE FOR USE WITH NORMAL WEIGHT HARDROCK CONCRETE AND GRADE 60 UNCOATED REINFORCING BARS. FOR LIGHTWEIGHT AGGREGATE USE 1.3ℓ
2. TOP BARS ARE HORIZONTAL BARS WITH 12" OR MORE OF CONCRETE CAST IN THE MEMBER BELOW THE BAR.
3. FOR BARS ENCLOSED IN STANDARD COL. SPIRALS, USE 0.75ℓ OR 12" MIN.
4. DEVELOPMENT LENGTH OF INDIVIDUAL BARS WITHIN A BUNDLE SHALL BE 1.2ℓ D FOR THAT BAR IN A 3–BAR BUNDLE AND 1.33ℓ D FOR A 4–BAR BUNDLE.
5. COMPRESSION DEVELOPMENT LENGTH (ONLY WHERE INDICATED ON DRAWINGS) FOR GRADE 60 BARS USE 22 BAR DIAMETERS.
6. CASE SELECTION
   –FOR FOUNDATION, SLAB ON GRADE, COLUMN REINFORCEMENT AND DOWELS, BEAM REINFORCEMENT, WALL REINFORCEMENT AND DOWELS (EXCEPT AS NOTED BELOW), AND WALLS WITH A SINGLE MAT OF STEEL CENTERED IN THE WALL AND DOWELS USE CASE 1 U.N.O.
   –FOR STRUCTURAL SLAB REINFORCEMENT AND CHORD STEEL REINFORCEMENT, USE CASE 2, U.N.O.

BAR SIZE	f'c = 3000 psi			
	TOP BARS		OTHER BARS	
	CASE 1	CASE 2	CASE 1	CASE 2
#3	19	28	15	22
#4	25	37	19	29
#5	31	47	24	36
#6	37	56	29	43
#7	54	81	42	63

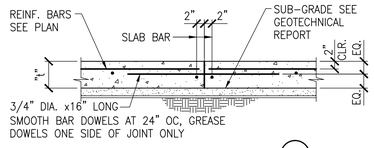


TENSION DEVELOPMENT LENGTH (CONCRETE ONLY) 2
NTS

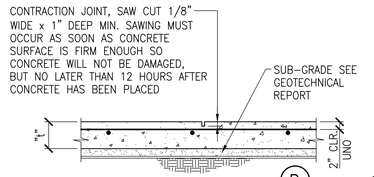


NOTES:
1. SLAB SHALL BE PLACED IN STRIP PATTERN.
   1 =FIRST 2 =SECOND
2. STRIPS TO BE DIVIDED BY CONSTRUCTION JOINTS AT THE CENTERLINE OF COLUMNS WHERE THEY OCCUR AND SUBDIVIDED AS REQUIRED INTO AREAS NOT EXCEEDING 225 SQ. FT. BY CONTRACTION JOINTS.
3. IN AREAS WHERE COLUMNS DO NOT OCCUR PROVIDE CONSTRUCTION AND CONTRACTION JOINTS AS ABOVE.
4. CONTRACTORS SHALL OBTAIN ARCHITECT'S APPROVAL FOR ALL JOINT LOCATIONS.

METHOD OF PLACING SLAB ON GRADE 3
NTS

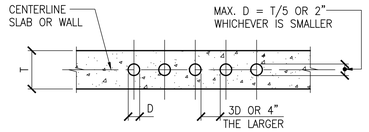


CONSTRUCTION JOINT A

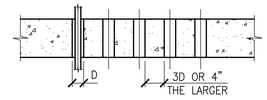


CONTRACTION JOINT B

PIPING CONDUIT IN OR THRU SLAB OR WALL 4
NTS

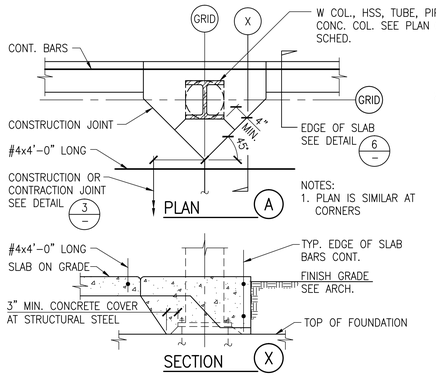


CONDUIT IN SLAB OR WALL

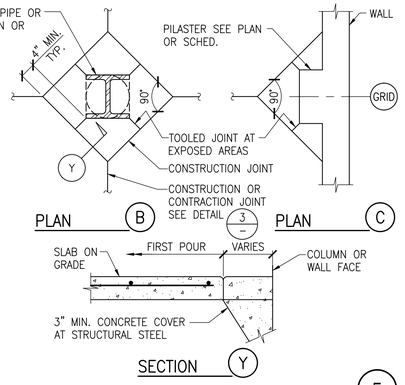


PIPING THRU SLAB OR WALL

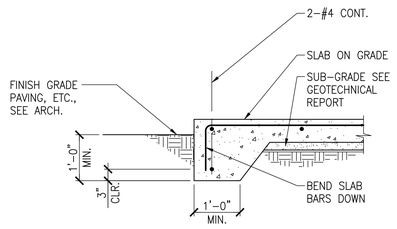
NOTES:
1. WHERE CLEAR DISTANCE BETWEEN SLEEVES IS IMPOSSIBLE THIS AREA SHALL BE TREATED AS A SLAB OPENING OR AS A WALL OPENING.



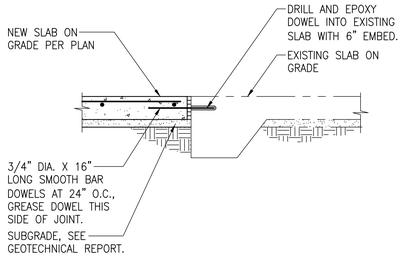
NOTES:
1. PLAN IS SIMILAR AT CORNERS



COLUMN ISOLATION JOINTS 5
NTS

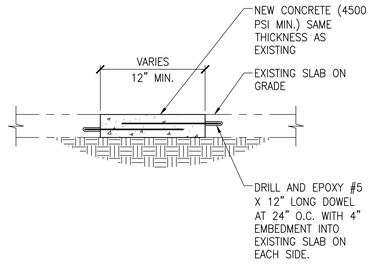


EDGE OF SLAB ON GRADE 6
NTS



NOTE:
1. DRILL AND EPOXY PER SECTION 'K' OF GENERAL STRUCTURAL NOTES ON SHEET S1.10

NEW SLAB AT EXISTING EDGE OF SLAB 7
NTS



NOTES:
1. PROVIDE #4 AT 24" O.C. IN DIRECTION WHERE OPENING EXCEEDS 24"
2. DRILL AND EPOXY PER SECTION 'K' OF GENERAL STRUCTURAL NOTES ON SHEET S1.10

EXISTING SLAB ON GRADE REPAIR 8
NTS

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1 REVISIONS 06/06/11
2 PERMIT REVIEW 06/27/11
3 PERMIT REVIEW 07/08/11

CONSTRUCTION DOCUMENTS

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KEY:

SHEET TITLE:
FOUNDATION DETAILS

SHEET NUMBER:
S5.01

DRAWN BY: EK	REVIEWED BY: SN/MT
DATE: 06/06/11	PROJECT NUMBER: 1101.00